Lecture 5-1: Regression with Two Regressors

Goal: Introduce mechanics of multiple regression with two regressors.

- Motivation for multiple regression
  - Richer description and prediction
  - Control for confounders
  - Model nonlinearity and interactions
  - Parameters, estimators, estimates
- Two cases
  - Binary + continuous
  - Continuous + continuous
- Mechanics of multiple regression
  - Derivation and Properties of OLS
  - Analysis of Variance
  - Partialling out interpretation

Worksheet

1. (*Interpreting a Binary Covariate in a Two–Regressor Model*) Suppose

$$\widehat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2, \qquad (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2) = (78.3,\ -10.6,\ -21.5),$$

where y is executive corruption (0–100), x_1 is high court independence (0–4), and $x_2 \in \{0, 1\}$ indicates democracy.

   (a) For a fixed level of court independence x_1, compute the predicted difference $\widehat{y}(x_2 = 1) - \widehat{y}(x_2 = 0)$ and interpret.

   (b) For a fixed regime type, compute the predicted change when x_1 increases by one unit.

   (c) Verify that the slope on court independence is the same in both groups and explain why.

2. (*Regression Plane and Counterfactual Prediction*) With two continuous covariates, suppose

$$\widehat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2, \qquad (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2) = (166.4,\ -13.95,\ -22.0),$$

where y is executive corruption (0–100), x_1 is high court independence (0–4), and x_2 is $\log_{10}$ GDP per capita.

(a) Show that for any two units A and B,

$$\widehat{y}_A - \widehat{y}_B = \hat{\beta}_1(x_{1A} - x_{1B}) + \hat{\beta}_2(x_{2A} - x_{2B}).$$

(b) Using $(x_1, x_2) = (2.96,\ 4.48)$ for Chile and $(0.21,\ 4.38)$ for China, compute the predicted levels and their difference.

(c) Explain the "partial effect" meaning of $\hat{\beta}_1$ and $\hat{\beta}_2$.

3. (*Properties of the fitted values and residuals*) Show that

(a) $\sum_{i=1}^{n} \hat{u}_i = 0 \Rightarrow \bar{y} = \hat{\bar{y}}$.

(b) The point $(\bar{y}, \bar{x}, \bar{z})$ is always on the regression plane.

(c) $Cov(x, \hat{u}) = Cov(z, \hat{u}) = 0 \Rightarrow Cov(\hat{y}, \hat{u}) = 0$.

4. (*ANOVA Decomposition, R^2, and Adjusted R^2*) Let $\hat{y}_i$ be fitted values from OLS with intercept and two regressors, and $\hat{u}_i = y_i - \hat{y}_i$. Prove that

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n}\hat{u}_i^2 \quad (\text{SST} = \text{SSE} + \text{SSR}).$$

Then explain why $R^2 = \text{SSE}/\text{SST}$ is weakly increasing when a regressor is added.